

The Spirit of Kangie

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Abstract

This paper shares the case study of *The Spirit of Kangie* which leveraged volumetric capture (vol-cap) to recontextualize dancing. The original project, *The Spirit of Obon*, extends choreographer Nobuko Miyamoto's original intent behind "Kangie" (meaning "gathering of joy") to bring people together after the COVID-19 pandemic. *Bon Odori* or Obon dance is a participatory circular dance performed communally during the Japanese and Japanese American Obon Festivals which commemorate our ancestors. At annual pilgrimages to Japanese American incarceration sites, *Bon Odori* is performed to remember Japanese Americans who were illegally incarcerated during World War II. Artist Nobuko Miyamoto's family was forced into relocation during WWII.

The vol-cap performances of Nobuko Miyamoto, Chie Saito, and Lani Yamanaka at Metastage Studio, a Microsoft Mixed Reality capture studio, transformed a traditional communal Obon dance into multiple distinct projects. The authors of this project built multiple archives of the artists' performances to optimize the original assets from *The Spirit of Obon* to evolve into *The Spirit of Kangie*, a microfilm, a portable 3D holographic display utilizing Looking Glass, and a wearable AR dance project with Qualcomm's Snapdragon AR glasses. This case study examines how 3D assets have facilitated a form of transplanted dance gathering when faced with considerable constraints - financial and environmental - while considering accessibility. Together, the authors of this paper alongside collaborators created experiences that highlight the profound importance of Japanese American community dancing, while offering insights into digital cultural preservation and interactive dance education with motion capture, holographic, wearable, and ubiquitous technologies.

CCS Concepts

• **Human-centered computing** → *Accessibility; Interaction devices*; • **Applied computing** → *Performing arts*;

Keywords

Holography, Real-time Human Interactive Computing, Education, Digital Cultural Preservation, Motion Tracking, AR Glasses, Looking Glass

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1 Volumetric Capture and Hologram

The Spirit of Kangie explores the use of real-time volumetric holography and motion tracking to support embodied cultural learning through interactive media. The installation featured life-sized holographic projections of Japanese folk dances on Hologauze from Holitronica, a fully silver projection gauze. Volumetric recordings of dancers demonstrating the steps were captured at Metastage in Los Angeles, using a 106-camera array (53 RGB and 53 infrared) surrounding a 35-foot diameter stage to support high-fidelity 3D reconstruction of the choreography for the holographic projections.



Figure 1: Obon hologram displayed at Georgia Tech Ferst Center for the Arts, October 2023 [2]

In the gallery, a custom interactive system used a Stereolabs camera array and a dedicated machine learning vision processor to generate real-time skeletal data that tracked multiple visitors simultaneously. This data informed responsive playback of holographic instructional sequences stored as textured 3D meshes, allowing the system to modulate pacing and presentation based on user motion. The custom software was integrated with TouchDesigner and Unity modules to support real-time holographic video playback, responsive interaction logic, and transitions between instructional states.



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The project demonstrates how volumetric holography and real-time interaction can facilitate culturally grounded, kinesthetic learning experiences. It also highlights the potential of combining motion capture, machine learning, and responsive media environments to support participatory engagement with traditional performance practices.

Future work might explore how holographic projection systems can evolve beyond instruction to support site-specific performance experiences that unfold dynamically as visitors learn the dances. By linking projected choreography to real-time visitor motion and spatial position, the system could transform the gallery into an adaptive performance environment, where the holograms respond to more than just the collective spatial composition of participants. Context-aware projection techniques, such as location-triggered visual elements or coordinated lighting changes, could be used to choreograph immersive group interactions. Enhancements to the body tracking pipeline could include expanded gesture recognition models capable of interpreting more nuanced, culturally specific movement vocabularies. Real-time volumetric rendering of participants could enable them to see themselves dancing within the holographic scene, reinforcing embodied feedback. Support for networked, multi-user configurations would allow geographically distributed participants to engage in shared learning environments. Adaptive instructional logic, informed by reinforcement learning, could personalize pacing and movement guidance over time. Augmenting the stereo vision processing with wearable inertial sensors may improve tracking reliability in variable lighting or crowded settings.

2 Wearable AR (Augmented Reality)

In addition to the real-time interactive elements of *The Spirit of Obon*, we place accessibility and affordability as highly important for engagement with all audiences. Through the means of AR (augmented reality) glasses that blend the physical world and virtual world, our hologram assets will be viewable, offering a more space conscious option for presenters and users. Unlike VR (virtual reality), which envelops viewers into a fully enclosed digital environment, AR uses the real world to enhance interactivity. Using tools like Unity, Kangie can be converted from our physical hologram to spatial computing assets to be loaded into AR/VR environments. In this case, through Qualcomm's Snapdragon AR ThinkReality Glasses, users view Chie Saito as an instructor of Kangie and can follow along in their own space. Users are able to follow Chie through viewing movements, listening to the music and instructions, and reading the words on the screen.

The practical application of AR glasses for those with accessibility needs is clear, allowing geographically distributed participants to engage in shared learning environments. Adaptive instructional logic improves in AR glasses with aspects such as accelerometers, reinforcement learning, and personalized guidance. The stereo vision processing with wearable inertial sensors expands low vision and hard of hearing accessibility and may also improve reliability in variable lighting or crowded settings such as galleries. AR can assist with visual overlays like closed captioning in multiple languages, and it can also provide a more clear listening option through the speakers close to one's ears. Furthermore, the use of AR glasses and



Figure 2: Qualcomm's Snapdragon AR ThinkReality Glasses [1]

their hands-free nature provides people with restricted mobility with the ability to view Kangie without needing to navigate the room containing the original large hologram version of *The Spirit of Obon*. While many AR glasses are still in Beta testing, and affordability is unknown, we have access to our own AR glasses, so we can provide this as a performance learning option.

Future development possibilities for *The Spirit of Kangie* include developing a multi-user AR or VR application allowing remote interaction between users. Viewers could be overlaid in the virtual space allowing them to see each other and dance together while being in two different physical locations. This would require each user to have cameras set up in a room to track where the user is and what they are doing. We see this used in existing VR multiplayer games such as VRChat, where people can virtually interact with each other through an avatar controlled by their own physical movements.

3 Portable 3D Holographic Display

The Looking Glass device uses a lightfield display to showcase 3D affected assets for viewers to see in multiple perspectives. To create the holographic effect, a lenticular lens is overlaid on a very high resolution pixel display, giving each pixel not only a color, but also a direction. The dance performance is then photographed (or, in the case of *The Spirit of Kangie*, rendered) from 48 different viewpoints. The results are then combined in software such that all 48 movies play at once, with each pixel on the screen pulled from the image which matches the view as seen from that pixel's direction.

The result is a three-dimensional image which can be viewed by multiple people at the same time, without the need for special glasses. The device used for this project is one made by the Looking Glass Factory, located in Brooklyn, NY.

Dance performances were captured as 3D polygonal meshes with texture maps, which were then imported into the Unity real-time development platform. The Looking Glass plug-in for Unity was used to define a volumetric space around each dancer, and render 48 separate movies, each from a slightly different viewpoint. The resulting 8x6 grid of moving images is known as a "quilt", and serves as the input to the Looking Glass device.

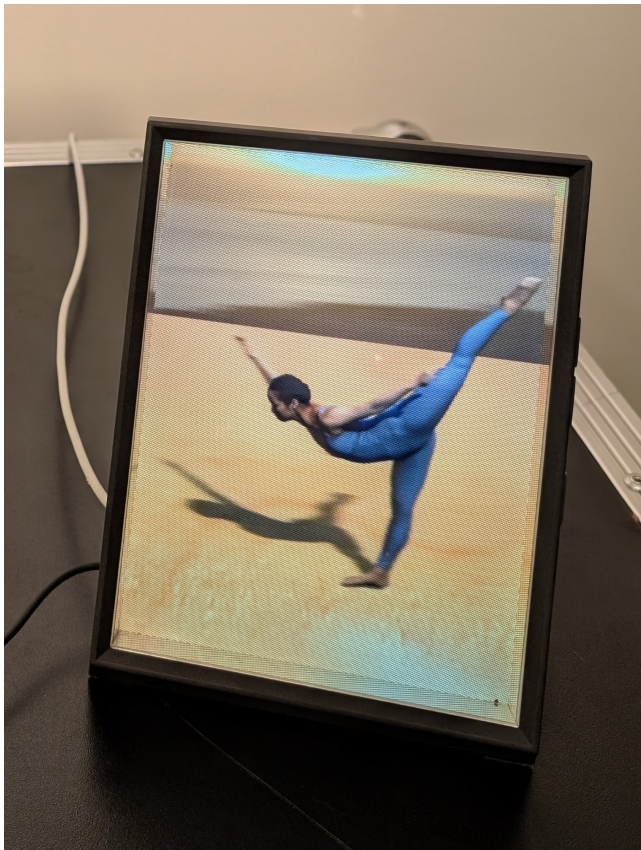


Figure 3: Looking Glass displayed at Japanese American Museum of San Jose, June 2025 [3]

4 Conclusion

Rooted in Buddhism, *Bon Odori* and Obon Festivals have become a broader cultural event that brings people together to share in community. *The Spirit of Obon* originated out of a larger extended reality exhibit on the illegal incarceration of Japanese Americans during World War II. The primary research was based on this prompt - when audiences are asked to interact through various extended reality programs to reflect on illegal Japanese American incarceration, would our empathy towards those whose civil liberties are

at risk change? Can technology help us understand belonging? The volumetric capture and holographic technology specifically applied to recontextualize and preserve a traditional dance experience transforms a site specific pilgrimage into an accessible journey. The Japanese American incarceration sites or war relocation centers are purposely scattered across the US in states including California, Colorado, Utah and Arkansas. To pilgrimage and dance at these sites is a trek many are unable to make. The new digital formatting of dancing Kangie increases exposure to the gathering of joy. The adaptability of the vol-cap assets improves practical constraints. When there is not enough space or environmental control for the holographic dance practice, we can easily switch to Snapdragon AR experience. Snapdragon AR glasses also have specialized audio for those hard of hearing. Low vision audiences may appreciate Looking Glass hand held brightening features. Anecdotally VR dance users have expressed the liberation of dancing in VR chat rooms as they reduce the barriers of entering a studio. These functional technologies add to the digital cultural preservation of Japanese American history and cultural dance education. Bridging traditional cultural practices with digital innovation focuses the *The Spirit of Kangie* - the gathering of joy - templating future applications and paths for similar interactive holographic systems with various hardware applications for cultural and educational contexts.

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- [3] Tiffany Schmidt. 2025. Photograph of Looking Glass at Japanese American Museum of San Jose.